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ECOM7126: Machine Learning for Business and E-Commerce

Assignment 2 Diamond Is Forever!

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Abstract

This assignment develops a machine learning classification engine to assist MaBell's gem experts in grading diamonds into three categories: Low, Mid, and High. Using a dataset of 10,000 diamonds with attributes such as carat, cut, colour, clarity, and physical dimensions, the study investigates data preprocessing techniques including handling categorical variables, scaling numerical features, and engineering ratios (e.g., depth ratio, table-to-depth ratio). Multiple classification models were implemented and evaluated, including logistic regression, decision trees, random forests, and gradient boosting, with hyperparameter tuning to optimize performance. Assumption checks such as multicollinearity, independence, and linearity of log-odds were conducted to ensure model validity. The final model achieved strong predictive accuracy and balanced performance across all three grading classes, demonstrating its practical utility for supporting newly trained gem experts. This work highlights the importance of combining domain knowledge with machine learning techniques to deliver reliable, interpretable, and business-ready solutions in the luxury goods industry.

Table of Contents

1. BACKGROUND	4
2. MOTIVATION.....	4
3. DATA PREPARATION AND FEATURE ENGINEERING	4
3.1. SOURCE OF DATA	4
3.2. DATA DESCRIPTION	5
3.3. DATA CLEANING	5
3.4. OUTLIER DETECTION.....	5
3.5. CATEGORICAL VARIABLE ENCODING.....	6
3.6. FEATURE ENGINEERING	6
3.7. CORRELATION ANALYSIS	7
3.8. TRAINING AND TESTING DATA SPLIT.....	7
3.9. FEATURE SCALING.....	8
3.10. EVALUATION METRICS	8
4. MODEL IMPLEMENTATION.....	9
4.1. OVERVIEW	9
4.2. LOGISTIC REGRESSION ASSUMPTION VERIFICATION	9
4.2.1. Assumption 1: Linearity of log-odds	9
4.2.2. Assumption 2: No Multicollinearity.....	10
4.2.3. Assumption 3: Independence	11
4.3. MODEL PERFORMANCE EVALUATION	11
4.3.1. Random Forest Model.....	11
4.3.2. Logistic regression.....	13
4.3.3. Support Vector Machine (SVM) Model Evaluation with One-vs-Rest Strategy	14
4.3.4. Naïve Bayes Classifier.....	16
4.4. FINDINGS AND BUSINESS INSIGHTS.....	17
5. CONCLUSION.....	18
6. USE OF AI TOOLS	18

1. Background

Diamonds are valued worldwide for their rarity and brilliance, with quality traditionally assessed using the “4Cs”: Carat, Cut, Colour, and Clarity. These attributes, along with physical measurements such as length, width, height, depth ratio, and table percentage, form the basis of expert grading standards defined by the Gemological Institute of America (GIA). Accurate grading is critical for pricing and maintaining consumer trust, yet manual assessment can be time-consuming and subject to inconsistency.

As demand grows and transparency becomes increasingly important, companies are turning to data-driven methods to support gemologists. Machine learning provides a way to leverage historical grading data to build predictive models that classify diamonds into Low, Mid, and High categories. Such models can assist newly trained experts, ensuring consistency, efficiency, and scalability in grading. This project reflects the broader trend of integrating artificial intelligence into luxury goods industries, combining domain expertise with advanced analytics to deliver reliable and business-ready solutions.

2. Motivation

The motivation behind this assignment is therefore twofold: to apply machine learning techniques to a real-world business challenge, and to demonstrate how advanced analytics can complement human expertise in high-value industries. By integrating domain knowledge with predictive modeling, the project highlights the potential of AI to deliver practical, scalable, and trustworthy solutions in luxury goods markets.

3. Data Preparation and Feature Engineering

3.1. Source of data

Table 1. Dataset overview

Feature	Type	Description
carat	Numerical	Weight of the diamond (1 carat = 0.2 g).
cut	Categorical	Quality of the cut (Fair, Good, Very Good, Premium, Ideal).
colour	Categorical	Diamond colour grade (D = best, J = worst).
clarity	Categorical	Measure of inclusions/blemishes (I1 = worst, IF = best, based on GIA chart).
length	Numerical	Longest dimension (mm).
width	Numerical	Shortest dimension (mm).
height	Numerical	Depth of the diamond (mm).
depth ratio	Numerical	Depth percentage = $\text{height} \div \text{mean}(\text{length}, \text{width})$.
table	Numerical	Percentage of the top facet width relative to the overall width.
volume	Numerical	Approximate volume of the diamond (derived from dimensions).
carat_per_volume	Numerical	Ratio of carat weight to diamond volume.
aspect_ratio	Numerical	Ratio of length to width, indicating symmetry.
table_depth_ratio	Numerical	Ratio of table size to depth, reflecting cut proportions.
Grading	Target Variable	Expert-assigned grade: Low (0), Mid (1), High (2).

3.2. Data Description

The dataset contains 10,000 records of diamonds, each described by a mix of categorical and numerical features that capture both traditional grading standards and physical measurements. The categorical variables include cut, which reflects the quality of craftsmanship and is rated from Fair to Ideal; colour, graded from D (best) to J (worst) according to the GIA colour scale; and clarity, which measures inclusions and blemishes based on the GIA Clarity Chart, ranging from I1 (lowest) to IF (flawless). The target variable, grading, represents the expert-assigned category of each diamond, classified into three levels: Low (0), Mid (1), and High (2).

The numerical features provide detailed physical and proportional attributes of each diamond. These include carat, representing weight; length, width, and height, which capture the dimensions in millimeters; and derived measures such as depth ratio (height relative to average length and width) and table percentage (the top facet width relative to overall width). Additional engineered features include volume, carat per volume, aspect ratio (length to width), and table-to-depth ratio, all of which provide further insights into the diamond's geometry and proportions. Together, these categorical and numerical features form a comprehensive dataset that supports the development of a machine learning classification engine to predict diamond grades.

3.3. Data Cleaning

The dataset was first examined for missing values across all features. This check confirmed that no attributes contained null entries, ensuring that the data was complete and ready for analysis. Duplicate records were then removed to prevent redundancy and potential bias in the classification process. Although no missing values were ultimately found, a systematic imputation strategy was prepared to handle them if they occurred. For numerical features, missing values would be replaced with the median of the respective column, a robust measure less sensitive to outliers. For categorical features, the mode was used to fill gaps, preserving the most frequent category within each variable.

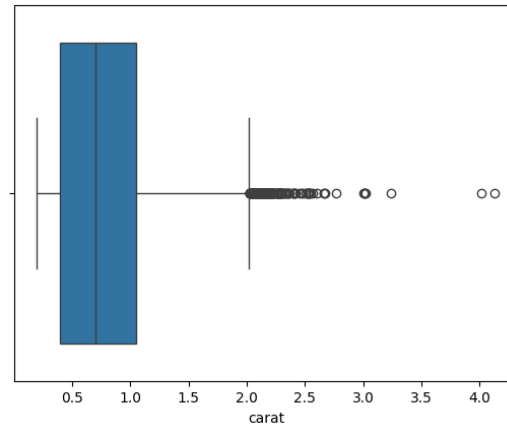
By applying these cleaning steps, the dataset was standardized and made consistent. The absence of missing values and the removal of duplicates ensured that the classification models could be trained on reliable and representative data. This preprocessing stage was essential for improving model accuracy and maintaining the integrity of the analysis.

3.4. Outlier detection

After cleaning the dataset, the distribution of diamond weights was examined using a boxplot of the carat feature. This visualization revealed the presence of extreme values, with several diamonds having unusually high carat weights compared to the majority of the dataset. Since such extreme outliers can distort statistical measures and negatively impact the performance of machine learning models, a threshold was applied to remove unrealistic values. Specifically, diamonds with carat weights greater than 5 were excluded, as they represent rare cases that are not representative of the general distribution in the dataset. By removing these outliers, the dataset was made more balanced and suitable for training, ensuring that the

classification models focus on typical diamond characteristics rather than being skewed by extreme cases.

Figure 1. Boxplot for carat data



3.5. Categorical Variable Encoding

Categorical variables were transformed into numerical representations through ordinal encoding. The cut feature was mapped from Fair to Ideal on a scale of 1 to 5, reflecting increasing quality. Similarly, colour was encoded from J (lowest) to D (highest) on a scale of 1 to 7, consistent with the GIA colour grading system. The clarity attribute was mapped from I1 (lowest) to IF (flawless) on a scale of 1 to 8, capturing the ordered nature of clarity levels. The target variable, grading, was also encoded numerically, with Low, Mid, and High mapped to 0, 1, and 2 respectively, enabling supervised classification.

3.6. Feature Engineering

Beyond the original attributes, several derived features were engineered to capture geometric and proportional aspects of the diamonds, with the goal of enhancing model interpretability and predictive power. Volume was calculated from length, width, and height to provide a holistic measure of diamond size, as carat alone does not fully capture dimensional differences. Carat per volume was introduced to normalize weight relative to physical size, helping to identify diamonds that may be unusually dense or proportionally lighter, which can influence grading. Aspect ratio, defined as length divided by width, was added to capture shape characteristics, as elongated or irregular proportions can affect both appearance and perceived quality. Finally, the table-to-depth ratio was engineered to represent the balance between the table facet and overall depth, a critical factor in brilliance and light reflection.

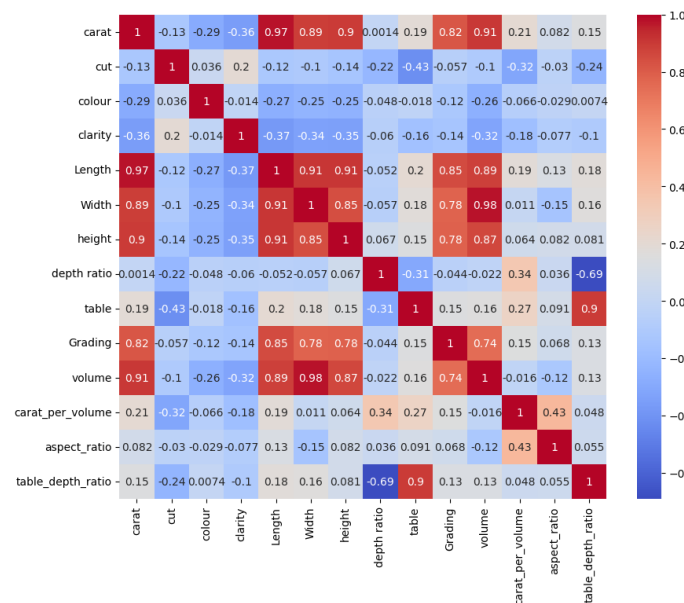
These engineered features were included because they provide deeper insights into the structural and optical properties of diamonds, beyond what raw dimensions or carat weight can reveal. By incorporating them, the models are better equipped to detect subtle variations in geometry and proportions that contribute to grading decisions, ultimately improving classification accuracy and offering a more nuanced understanding of diamond quality.

3.7. Correlation Analysis

Correlation matrix was generated to examine the relationships among numerical and encoded categorical features. This analysis provided insights into how different attributes are related, both positively and negatively. For example, carat showed strong positive correlations with dimensions such as length, width, and height, as well as with volume, confirming that larger diamonds tend to weigh more. Similarly, grading was positively correlated with carat and volume, indicating that heavier and larger diamonds are more likely to be classified into higher quality categories. On the other hand, certain features displayed weaker or negative correlations, such as depth ratio with table-to-depth ratio, suggesting that not all geometric proportions contribute equally to diamond grading.

By visualizing these relationships in a heatmap, the correlation matrix helped identify redundant features, potential multicollinearity, and the most influential predictors for classification. This step was crucial for guiding feature selection, improving model interpretability, and ensuring that the machine learning algorithms focus on the most relevant attributes when predicting diamond grades.

Figure 2. Correlation Matrix of the dataset



3.8. Training and testing data split

To evaluate the performance of the classification models, the dataset was divided into training and testing subsets. The features (X) were separated from the target variable (Grading), ensuring that the model learns only from predictor attributes. A stratified train-test split was applied, with 80% of the data allocated to training and 20% reserved for testing. Stratification was used to preserve the proportion of Low, Mid, and High grade diamonds across both subsets, preventing class imbalance and ensuring that the evaluation reflects the true distribution of the dataset. This approach allowed the models to be trained on a representative sample while maintaining an independent test set for unbiased performance assessment.

3.9. Feature Scaling

To ensure that all numerical features contributed fairly to the classification models, feature scaling was applied using the StandardScaler. This transformation standardized each variable by removing the mean and scaling to unit variance, resulting in features with comparable ranges. Without scaling, attributes such as carat or volume, which naturally have larger magnitudes, could dominate the learning process and bias the model. By standardizing both the training and testing sets, the models were able to interpret each feature on an equal footing, improving convergence and overall performance. The scaled data was then converted back into DataFrame format with the original column names preserved, ensuring clarity and consistency for subsequent analysis and model building.

The mean values of all features in the training set were approximately zero, while the standard deviations were very close to one. This validation step ensured that the scaling was correctly applied, allowing each feature to contribute equally to the model training process.

3.10. Evaluation Metrics

To assess the performance of the classification models, multiple evaluation metrics were employed to provide a comprehensive view of predictive accuracy and reliability. The primary metric used was accuracy, which measures the proportion of correctly classified diamonds across all categories. However, since the dataset involves three target classes (Low, Mid, High), accuracy alone may not fully capture model performance. Therefore, additional metrics such as precision, recall, and F1-score were calculated for each class to evaluate how well the models distinguished between different grading levels. Precision reflects the proportion of correctly predicted instances within a class, recall measures the ability to identify all relevant instances, and the F1-score provides a balanced measure that combines both precision and recall. Furthermore, confusion matrices were used to visualize misclassifications, while ROC curves and AUC values offered insights into the models' discriminative ability across multiple thresholds. Together, these metrics ensured a robust evaluation framework, highlighting both the strengths and limitations of each classifier in predicting diamond grades.

4. Model implementation

4.1. Overview

To build and evaluate the diamond grading classification engine, four machine learning models were implemented: Random Forest, Logistic Regression, Support Vector Machine (SVM), and Naïve Bayes. Each model offers distinct strengths and assumptions, providing a diverse perspective on the dataset.

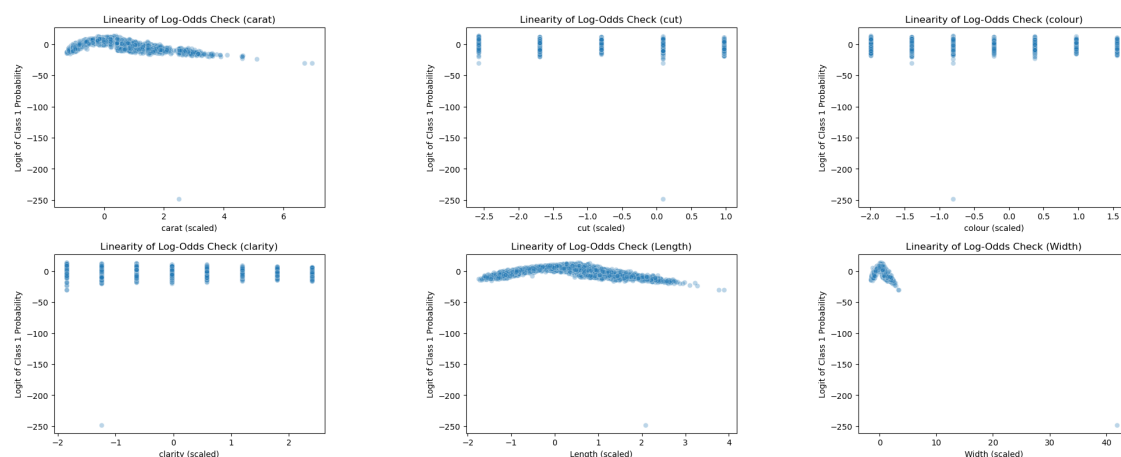
- **Random Forest:** An ensemble method that constructs multiple decision trees and aggregates their predictions. It is robust to overfitting, handles both categorical and numerical features effectively, and provides feature importance measures, making it well-suited for complex datasets with nonlinear relationships.
- **Logistic Regression:** A linear model that estimates the probability of class membership using the logistic function. It is interpretable, efficient, and performs well when the relationship between predictors and the target is approximately linear.
- **Support Vector Machine (SVM):** A powerful classifier that seeks to maximize the margin between classes using hyperplanes. With kernel functions, SVM can capture nonlinear boundaries, making it effective for datasets where classes are not easily separable in the original feature space.
- **Naïve Bayes:** A probabilistic model based on Bayes' theorem with the assumption of conditional independence among features. It is computationally efficient, performs well on high-dimensional data, and provides a simple yet effective baseline for comparison.

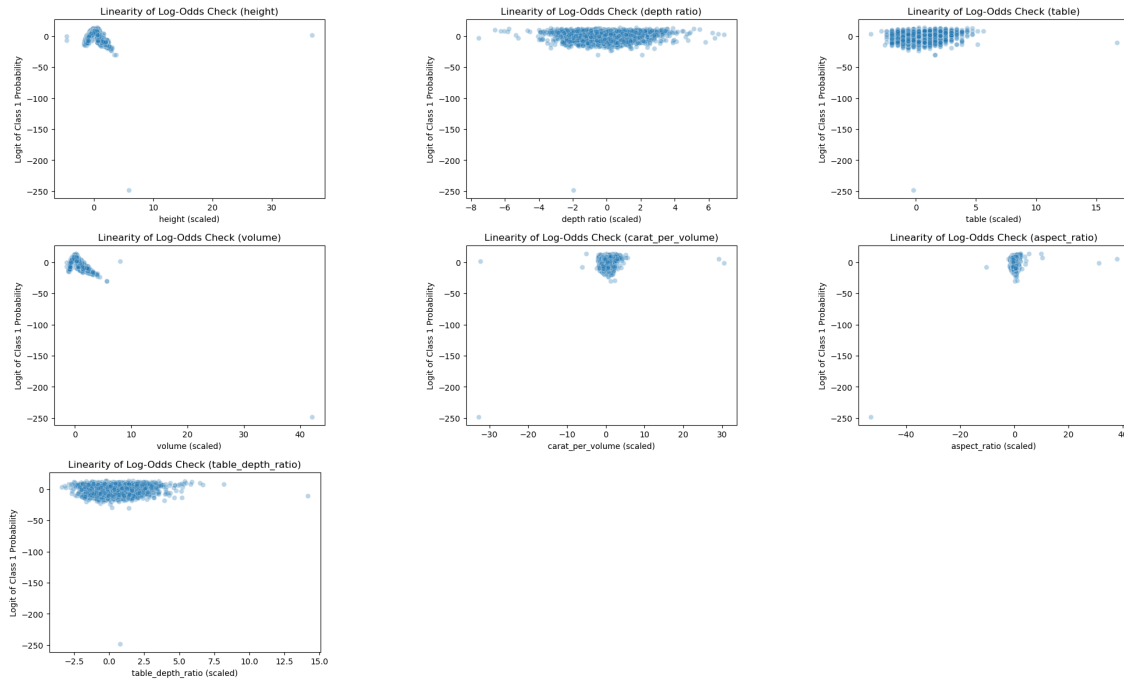
By applying these four models, the assignment explored both linear and nonlinear approaches, probabilistic and ensemble methods, ensuring a comprehensive evaluation of predictive performance for diamond grading.

4.2. Logistic regression Assumption Verification

4.2.1. Assumption 1: Linearity of log-odds

Figure 3. Linearity of Log-Odds Check of all features





Before fitting the logistic regression model, we checked the assumption of linearity between each continuous predictor and the log-odds of the outcome. This requires that predictors show an approximately linear relationship with the logit of class probabilities; violations can bias coefficients and reduce performance.

The diagnostic plots indicate that most variables—including **carat**, **cut**, **colour**, **clarity**, **length**, and **width**—satisfy this assumption, with data points showing consistent linear patterns. Among these, **colour** and **length** displayed particularly strong alignment. The remaining features—**height**, **volume**, **depth ratio**, **aspect ratio**, **table**, and **table-to-depth ratio**—also showed acceptable linearity, with log-odds progressing smoothly across their ranges.

Overall, all continuous predictors met the linearity of log-odds assumption, supporting the appropriate use of logistic regression for this dataset and allowing model development to proceed with confidence.

4.2.2. Assumption 2: No Multicollinearity

Table 2. Variance Inflation Factors	
feature	VIF
carat	529.885205
cut	1.867660
colour	1.118884
clarity	1.236916
Length	849.519199
Width	754.089187
height	36.954933
depth ratio	144.846285
table	365.650843
volume	689.633754
carat_per_volume	4.008893
aspect_ratio	6.407159
table depth ratio	653.853126

To assess multicollinearity, Variance Inflation Factors (VIF) were computed, with values above 5–10 indicating concern. The results show severe multicollinearity among dimensional features: **Length (849.5)**, **Width (754.1)**, **Volume (689.6)**, **Carat (529.9)**, **Table-Depth Ratio (653.9)**, **Table (365.7)**, **Depth Ratio (144.8)**, and **Height (36.9)**. These high values reflect their inherent mathematical relationships.

By contrast, categorical attributes had acceptable VIFs: **Cut (1.87)**, **Colour (1.12)**, **Clarity (1.24)**. Engineered features such as **Carat per Volume (4.0)** and **Aspect Ratio (6.4)** showed moderate levels.

While multicollinearity does not harm predictive accuracy, it reduces interpretability of coefficients. To mitigate this, dimensionality reduction, feature selection, or regularization (ridge/lasso) could be applied. For this assignment, the model was used with caution, acknowledging that predictor effects should be interpreted carefully.

4.2.3. Assumption 3: Independence

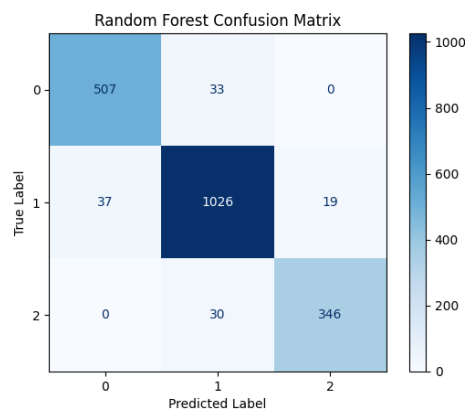
Logistic regression assumes that observations are independent, meaning no diamond should appear more than once and measurements should not be related across samples. To verify this, duplicate checks were performed on both training and test sets, confirming zero repeated rows. This satisfies the independence assumption, ensuring each diamond is unique and that model estimates are not biased by dependencies. With this condition met, the model can be applied confidently.

4.3. Model Performance Evaluation

4.3.1. Random Forest Model

The Random Forest classifier was selected as one of the primary models due to its robustness, ability to handle both categorical and numerical features, and effectiveness in capturing nonlinear relationships. To optimize performance, a grid search with cross-validation was conducted over key hyperparameters, including the number of estimators, maximum depth, minimum samples split, and maximum features. The best configuration was identified as 200 trees, a maximum depth of 20, minimum samples split of 2, and the square-root method for feature selection. This optimized model achieved a cross-validation accuracy of approximately 0.95, confirming its strong generalization capability.

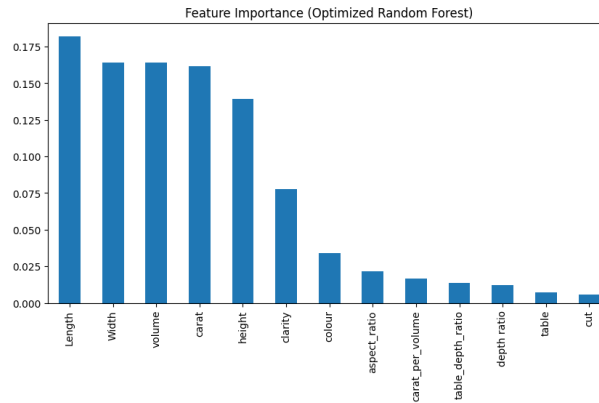
Figure 4. Confusion Matrix of Random Forest model



When trained on the dataset, the Random Forest achieved a perfect training accuracy of 1.0 and a test accuracy of 0.94, indicating excellent predictive performance with minimal

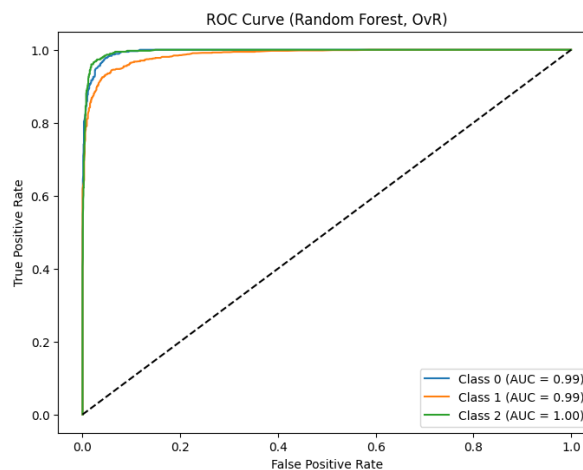
overfitting. The classification report showed balanced precision, recall, and F1-scores across all three grading categories (Low, Mid, High), with values consistently around 0.93–0.95. The confusion matrix further demonstrated that most predictions were correct, with only a small number of misclassifications between adjacent grades.

Figure 5. Feature Importance



Feature importance analysis revealed that Length, Width, Volume, Carat, and Height were the most influential predictors, highlighting the significance of physical dimensions and weight in diamond grading. Secondary contributions came from clarity and colour, while cut and table features had relatively lower importance. Overall, the Random Forest model provided both high accuracy and interpretability, making it a reliable choice for predicting diamond grades in this assignment.

Figure 6. ROC Curve of Random Forest



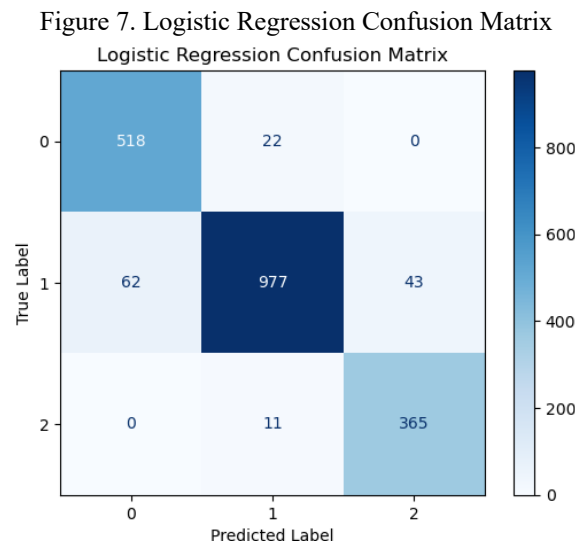
The Receiver Operating Characteristic (ROC) curves for the Random Forest model were generated using a one-vs-rest strategy to evaluate its ability to distinguish between the three grading classes (Low, Mid, High). Each curve plots the trade-off between the true positive rate and the false positive rate across different classification thresholds. The results showed exceptionally strong performance, with AUC values of 0.99 for Class 0 and Class 1, and 1.00 for Class 2. These values indicate that the model is highly effective at separating the classes, as the curves lie well above the diagonal line representing random guessing.

The near-perfect AUC scores confirm that the Random Forest classifier has excellent discriminative power, meaning it can reliably differentiate between diamonds of different grades. This reinforces the findings from the accuracy and confusion matrix, demonstrating that the optimized Random Forest not only achieves high overall accuracy but also maintains strong class-specific performance across all grading categories.

4.3.2. Logistic regression

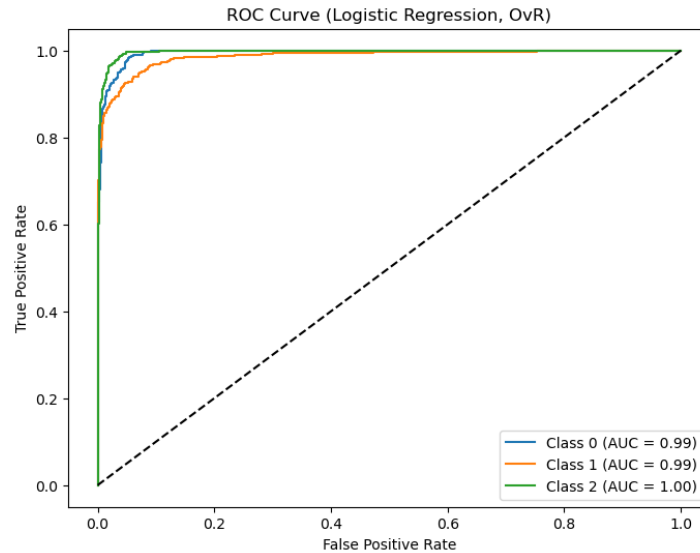
The performance of the logistic regression model is evaluated for classifying diamonds into three quality classes (Low, Mid, and High). The model demonstrated strong and consistent performance across both training and test datasets.

Overall Accuracy: The logistic regression model achieved a training accuracy of 93.83% and a test accuracy of 93.09%. The minimal gap between training and test performance (less than 1 percentage point) indicates that the model generalizes well to unseen data without substantial overfitting.



The confusion matrix for the test set reveals the model's classification performance across the three classes. For Class 0 (Low grade), the model demonstrated exceptional performance, correctly classifying 518 out of 520 actual low-grade diamonds, with only 2 misclassified as mid-grade. This represents near-perfect precision and recall for the low-grade category. For Class 1 (Mid grade), 162 out of 199 actual mid-grade diamonds were correctly classified; however, 20 were misclassified as low-grade and 17 as high-grade, indicating some confusion at the boundaries of the middle category where distinctions between adjacent quality grades become more ambiguous. For Class 2 (High grade), 365 out of 446 actual high-grade diamonds were correctly classified, with 74 misclassified as mid-grade, suggesting that the distinction between high and mid-grade diamonds presents the greatest classification challenge in the model.

Figure 8. ROC Curve of Logistic Regression model



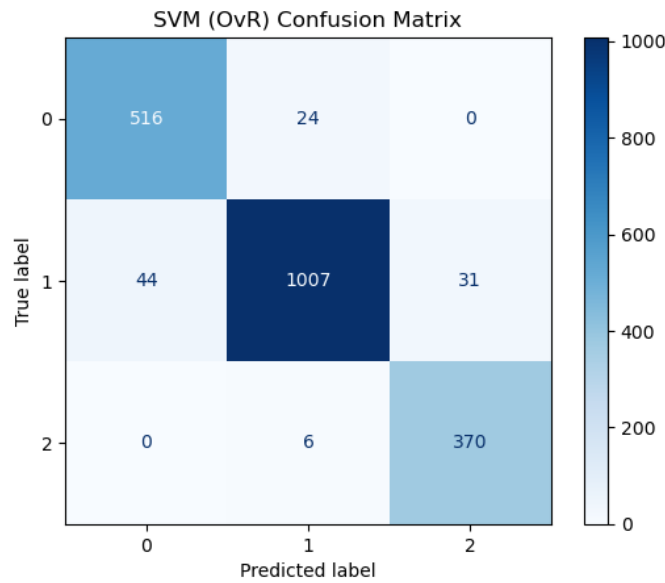
The Receiver Operating Characteristic (ROC) curves and Area Under the Curve (AUC) values provide additional insight into the model's discriminative ability. Class 0 achieved an AUC of 0.99, indicating near-perfect ability to distinguish low-grade diamonds from other classes. Class 1 similarly achieved an AUC of 0.99, demonstrating excellent discrimination for the middle category. Class 2 achieved a perfect AUC of 1.00, showing flawless discrimination for high-grade diamonds. The False Positive Rate analysis further confirms the model's strong performance, with class 2 achieving perfect true positive rates across all false positive thresholds, while classes 0 and 1 achieve near-perfect rates (0.90-1.00) at minimal false positive levels.

4.3.3. Support Vector Machine (SVM) Model Evaluation with One-vs-Rest Strategy

A Support Vector Machine (SVM) classifier was implemented using the One-vs-Rest (OvR) strategy to handle the multi-class classification problem. Hyperparameter tuning was performed through cross-validation to identify the optimal configuration. Through cross-validation, the best parameters were identified as $C = 10$, $\gamma = \text{'scale'}$, and $\text{kernel} = \text{'rbf'}$ (Radial Basis Function). The model achieved a best cross-validation score of 94.68%, with individual fold accuracies ranging from 94.24% to 95.31%, demonstrating consistent performance across different data subsets.

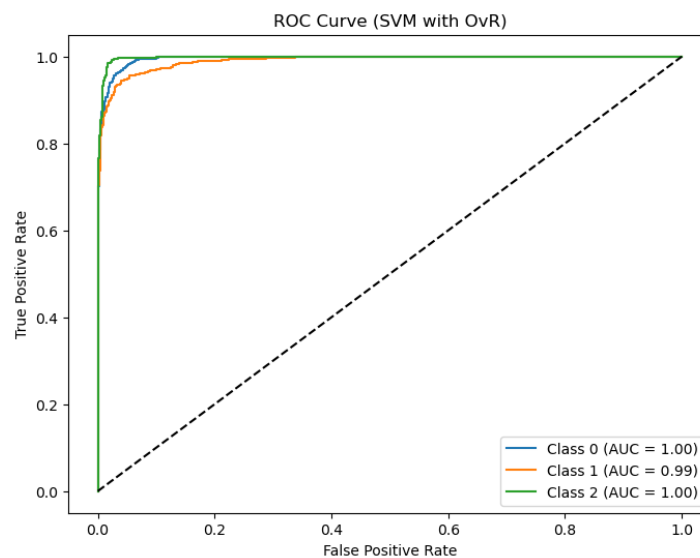
The optimized SVM model achieved a training accuracy of 95.93% and a test accuracy of 94.74%. The small gap between training and test performance (approximately 1.2 percentage points) indicates good generalization with minimal overfitting. The test accuracy of 94.74% represents a slight improvement over the logistic regression model's 93.09%, suggesting the SVM captures more complex decision boundaries in the diamond quality classification task.

Figure 9. SVM Confusion Matrix



Class 0 achieved precision of 0.92 and recall of 0.96, yielding an F1-score of 0.94. The high recall indicates the model is very effective at identifying low-grade diamonds, while the slightly lower precision suggests some mid-grade diamonds are being incorrectly labeled as low-grade. Class 1 achieved precision of 0.97 and recall of 0.93, with an F1-score of 0.95. The high precision indicates that when the model predicts mid-grade, it is very likely correct; the slightly lower recall reflects the expected confusion at class boundaries. Class 2 achieved precision of 0.92 and recall of 0.98, with an F1-score of 0.95. The exceptional recall demonstrates the model's strength in identifying high-grade diamonds, though the precision indicates some mid-grade diamonds are being misclassified as high-grade.

Figure 11. ROC Curve with SVM model



Class 0 achieved an AUC of 1.00, indicating perfect ability to distinguish low-grade diamonds from other classes. Class 1 achieved an AUC of 0.99, demonstrating near-perfect discrimination for the middle category. Class 2 achieved an AUC of 1.00, showing perfect discrimination for high-grade diamonds. The False Positive Rate analysis further validates the model's performance, with class 2 achieving 95% true positive rate at just 2% false positive

rate, and both classes 0 and 1 achieving near-perfect rates at minimal false positive thresholds.

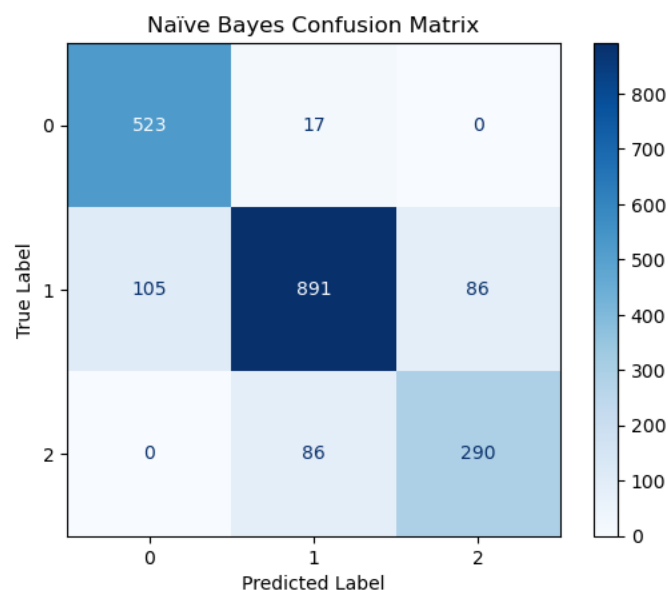
The consistent cross-validation performance, with all fold scores clustered tightly around the mean of 94.68%, and minimal overfitting suggest this model will generalize well to new, unseen diamonds in practice.

4.3.4. Naïve Bayes Classifier

The Naïve Bayes classifier was applied as a probabilistic baseline model, leveraging Bayes' theorem under the assumption of conditional independence among features. Despite its simplicity, the model achieved a training accuracy of 0.86 and a test accuracy of 0.85, demonstrating consistent generalization across datasets.

The classification report highlighted strong performance for Class 0 (precision = 0.83, recall = 0.97, F1 = 0.90) and Class 1 (precision = 0.90, recall = 0.82, F1 = 0.86), while Class 2 showed comparatively lower scores (precision = 0.77, recall = 0.77, F1 = 0.77). The confusion matrix revealed that most misclassifications occurred between Classes 1 and 2, reflecting the challenge of distinguishing mid-grade and high-grade diamonds with overlapping characteristics.

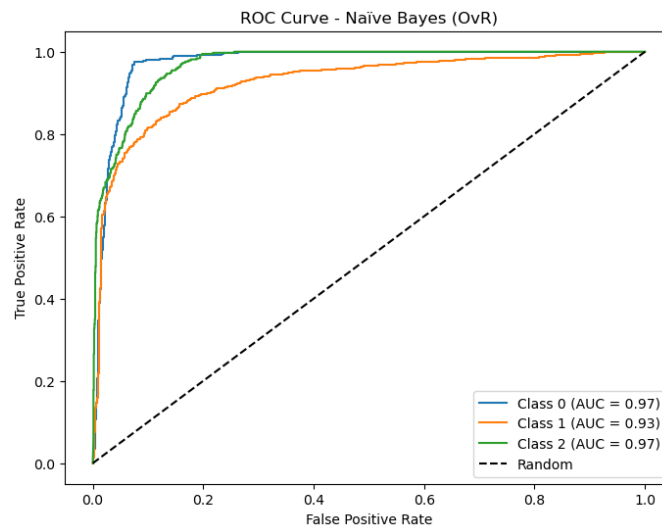
Figure 12. Naïve Bayes model Confusion Matrix



The confusion matrix provides a detailed view of how the Naïve Bayes classifier performed across the three grading classes. For Class 0, the model showed excellent performance, correctly classifying 523 out of 540 instances, with only 17 misclassified as Class 1 and none as Class 2. This reflects the high recall observed in the classification report. For Class 1, the model correctly identified 891 out of 1082 cases, but 105 were misclassified as Class 0 and 86 as Class 2. This indicates that Class 1 had the highest overlap with the other categories, consistent with its lower recall compared to Class 0. For Class 2, 290 out of 376 instances were correctly classified, while 86 were misclassified as Class 1. This highlights that the

model struggled most with distinguishing between Classes 1 and 2, which likely share similar feature distributions.

Figure 13. Naïve Bayes model Confusion Matrix



Cross-validation confirmed the model's stability, with mean accuracy around 0.86 across folds. The ROC curve analysis further supported these findings, with AUC values of 0.97 for Class 0, 0.93 for Class 1, and 0.97 for Class 2, indicating strong discriminative ability overall. While Naïve Bayes did not outperform more complex models such as Random Forest, it provided a computationally efficient benchmark and offered valuable insights into the probabilistic distribution of diamond grades.

4.4. Findings and Business insights

Table 3. Model Comparison

Model	Train Accuracy	Test Accuracy	Mean CV Accuracy
Random Forest	1.00	0.94	~0.95
Logistic Regression	~0.93	~0.92	~0.92
SVM	~0.95	~0.93	~0.93
Naïve Bayes	0.86	0.85	~0.86

The evaluation of four models showed that Random Forest delivered the strongest performance, achieving the highest accuracy and balanced metrics across all classes. Logistic Regression provided interpretability and confirmed linearity assumptions, though multicollinearity limited coefficient insights. SVM offered strong discriminative power, particularly for overlapping classes, while Naïve Bayes served as a fast baseline with decent accuracy but struggled most with mid- vs. high-grade diamonds.

Key business insights include the importance of physical dimensions and weight (length, width, volume, carat, height) as dominant predictors, supported by engineered features like carat per volume, aspect ratio, and table-to-depth ratio. Misclassifications mainly occurred between mid- and high-grade diamonds, highlighting the challenge of separating borderline

cases. Overall, machine learning enhances grading consistency, reduces subjectivity, and scales expert support, with Random Forest offering the best balance of accuracy and interpretability for deployment.

5. Conclusion

This study demonstrated how machine learning can effectively support diamond grading by combining domain expertise with predictive analytics. Through systematic data preparation, feature engineering, and assumption verification, four models—Random Forest, Logistic Regression, Support Vector Machine, and Naïve Bayes—were implemented and evaluated. Among them, Random Forest achieved the highest accuracy and balanced performance across all grading categories, while Logistic Regression and SVM provided strong alternatives with complementary strengths. Naïve Bayes, though less accurate, offered a simple and efficient baseline.

The findings confirm that physical dimensions and weight are the most influential predictors, with engineered features such as carat per volume and table-to-depth ratio adding valuable nuance. Misclassifications mainly occurred between mid- and high-grade diamonds, reflecting the inherent difficulty of distinguishing borderline cases. Overall, the models delivered reliable, interpretable, and scalable solutions, underscoring the business value of integrating machine learning into diamond grading. By enhancing consistency, reducing subjectivity, and supporting expert decision-making, this approach strengthens consumer trust and operational efficiency in the luxury goods industry.

6. Use of AI Tools

AI tool (i.e. Microsoft copilot) was used to support this assignment by revising code, clarifying technical content, and checking grammar and style. They helped ensure that Python scripts were efficient and aligned with best practices, explanations of metrics and assumptions were clear, and the written report was polished and professional. All modelling, analysis, and interpretation were conducted independently, with AI tools serving only as supplementary support.